Risk Assessment & Mitigation

DEPI · Project Planning & Management · Railway Ticket Sales & Journey Performance

1. Purpose and scope

This section lists the risks that could delay the project or weaken its results, and how the team plans to handle each one. The data risks are based on a review of the dataset (`railway.csv`: 31,653 ticket records, 18 columns, journeys from 1 Jan to 30 Apr 2024) and its data dictionary. Project and tool risks are based on the planned team structure and workflow (Python → Power Query → SQL → Power BI).

2. Summary

10	5	4	1	4
Total risks	Data	Project	Tools	High impact

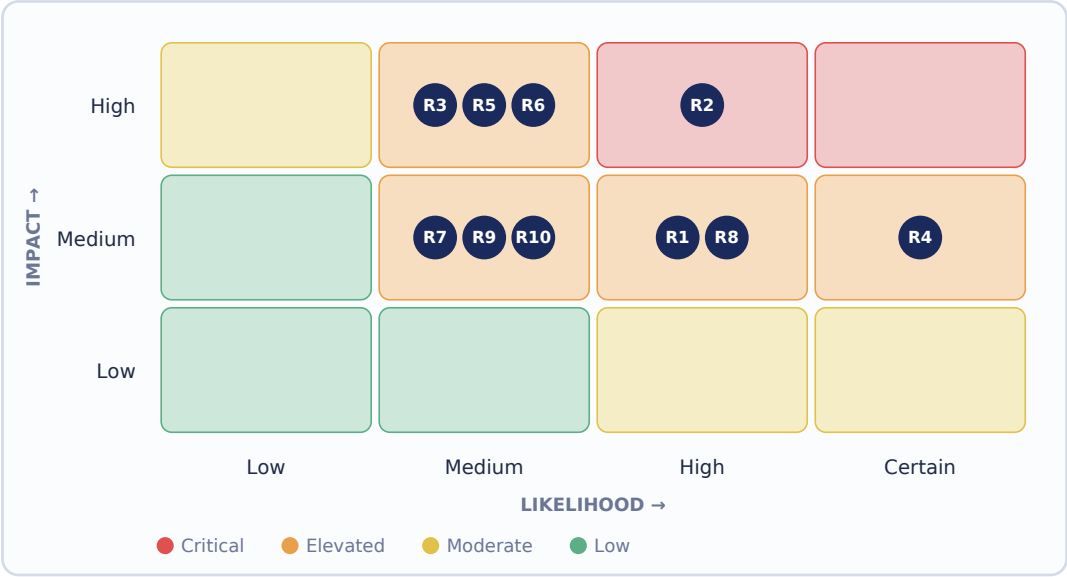
3. Rating scale

- **Likelihood:** Certain (already true) · High · Medium · Low.
- **Impact:** High (blocks or invalidates results) · Medium (slows work or limits conclusions) · Low (minor).

4. Notes

- Some data checks found no problems: no duplicate rows, no duplicate Transaction IDs, and no journey starting and ending at the same station.
- Blank values in `Actual Arrival Time` (cancelled journeys) and `Reason for Delay` (on-time journeys) are expected and must not be deleted or filled.
- The machine learning model is optional (stretch goal). It is tracked as R10 and does not change the required deliverables.
- This register will be reviewed when the team finalizes any additional data to be added.

Risk heat map



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5. Risk register

ID	Risk	Type	Likelihood	Impact	Mitigation
R1	Same value written in different ways in Reason for Delay (e.g. Signal Failure / Signal failure, Weather / Weather Conditions), which splits the totals.	**Data**	High	Medium	Standardize the values during the Cleaning stage and record the rule in the cleaning rules file.
R2	Date and time columns are stored as text, and some journeys arrive after midnight, so delay duration can be calculated wrongly.	**Data**	High	**High**	Convert data types in Power Query and calculate delay using the arrival date as well as the time.
R3	The value None in Railcard (20,918 rows) may be read as a missing value (Null) by some tools, losing its meaning of "no railcard".	**Data**	Medium	**High**	Set the file-reading options to keep the value, and re-check the column after each stage.
R4	The original dataset covers only 4 months of journeys (Jan–Apr 2024), so seasonal trends cannot be concluded from it alone.	**Data**	Certain	Medium	State the limitation in the Scope and rely on monthly and weekly analysis unless additional data is added.
R5	The same KPI may be calculated differently between the Analysis and Visualization stages (e.g. refund request rate).	**Project**	Medium	**High**	Define each KPI once in a shared document before any calculation starts.
R6	The M Language Transformation stage depends on a single team member (who also reviews the SQL analysis), so a delay there delays the Analysis and Visualization stages.	**Project**	Medium	**High**	Document the steps, train another member as backup, and hold a short weekly check-in.
R7	Several members working on the Cleaning stage may make conflicting changes.	**Project**	Medium	Medium	Split the columns between members and keep one shared script on GitHub.
R8	Power BI (.pbix) files cannot be merged automatically when more than one member edits the dashboard.	**Tools**	High	Medium	One member owns the master file, pages are divided between members, and versions are numbered (v1, v2).
R9	Added data (columns or rows) may not match the original dataset in station names, dates or format, or may come from an unreliable source.	**Data**	Medium	Medium	Document each source and its join key, test the match rate before use, and mark added data as separate from the original.
R10	The optional machine learning model (stretch goal) may take time and attention away from the core work (SQL analysis, dashboard, testing).	**Project**	Medium	Medium	Decide only after the SQL analysis is completed, start only if the core milestones are on track, and drop it without affecting any required deliverable if time is short.